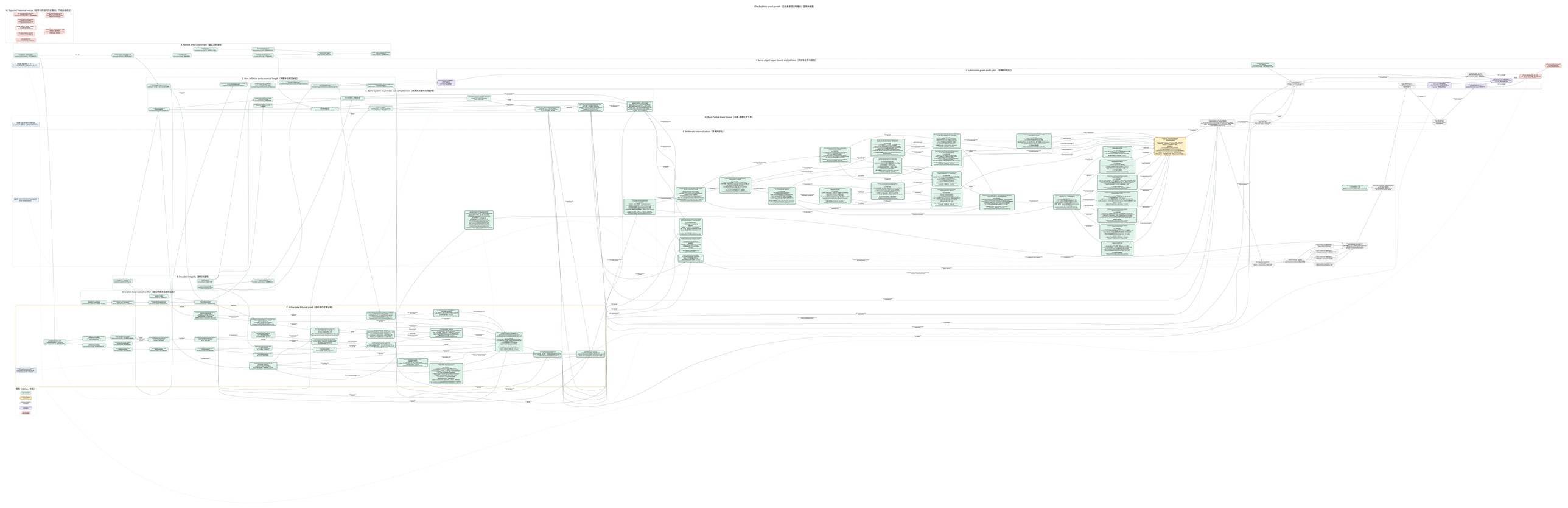

# Checked min-proof growth（已检查最短证明增长）定理依赖图

K. Rejected historical routes（经审计弃用的历史路线；不通向主结论）
Nat.size(minProofCode) lower bound
Prefix checker rebuilding rawCode
Unproved Hilbert-system simulation
strengthened_to_partial / transfer_to_graft
tail_gap / computable_gap input
upper_provider input
A. Honest proof coordinate（诚实证明坐标）
binaryNatCode / decodeBinaryNat
binaryTermCode + binaryFormulaCode
binarySequentCode
PAAxiomCertificate
StructuralValidityCertificate
CheckedPAProofTree binaryCode
binaryCertifiedPAProofCode
packBinaryString / packedPayloadLength
C. Non-inflation and canonical length（不膨胀与规范长度）
shift/freeVarg /substitution size bounds
decodeCompactTerm/Formula/Sequent_canonical_length_le
decodeCompactProof_parseWeight_le
decodeCompactProof_binaryLength_le
decodeStructuralValidityCertificate_binaryLength_le
compactCertifiedPAProofVerifier_canonical_length_bound
#print axioms audit（公理依赖审计）
E. Same-system soundness and completeness（同系统可靠性与完备性）
decodeCompactListedProof
decodeCompactListedProof_toChecked
listedCertificateValidTrace_toChecked_iff
listedCompactCertifiedPAProofVerifier_eq_true_iff
decodeCompactListedProof_canonicalBinaryCode_append
derivation_to_listedCompactCertifiedPAProofVerifier completeness（完备性）
listedCompactCertifiedPAProofVerifier_toDerivation soundness（可靠性）
Concrete P(x,y) / ListedCompactCertifiedPAProofChecks
minListedCertifiedPAProofPayloadLength(formula)
P_canon(code, formulaCode) = canonical Bool verifier
I. Same-object upper bound and collision（同对象上界与碰撞）
Sondow checked certificates
Sondow same-object compiler
J. Submission-grade audit gates（投稿级审计门）
minProofLength(F_n) <= U(n)
Final dependency audit（最终依赖审计）
U(N) < minProofLength(F_N) <= U(N)
Forbidden project assumptions（禁止项目级假设）
Pudlak 1986 Theorem 3.1
minProofLength(F_n) > U(n) eventually
H. Buss-Pudlak lower bound（布斯-普德拉克下界）
CompactProof(x,y) = Con_PA^compact(k)
F_n = Con_T(Z(n)^c)
Buss 1994 Theorem 5
G. Arithmetic internalization（算术内部化）
Quantitative parameterized diagonalization
Primitive-recursive packed input layer
Primitive-recursive binary-Nat token stream machine
Primitive-recursive arithmetic term/formula parser
Exact arbitrary-input syntax inversion
Primitive-recursive closed-formula token machine
Exact noncanonical bit token stream synchronization
Primitive-recursive PA/structural certificate parsers
Typed syntax decoder iff exact token prefix
Primitive-recursive listed-proof structural parser
PA/structural certificate typed decoder iff exact token prefix
Exact arbitrary-input PA/structural certificate inversion
Listed proof typed decoder iff exact token prefix
Exact arbitrary-input listed proof inversion
Primitive-recursive packed stream + exact typed-decoder equivalence
Primitive-recursive fixed PA axiom sentence table
Primitive-recursive term/formula/sequent value parsers
Primitive-recursive direct formula constructors
Primitive-recursive numeric formula-list checks
Primitive-recursive formula-fvd/r token machine
Primitive-recursive one-variable substitution token machine
Primitive-recursive formula free token machine
Primitive-recursive formula-shift token machine
Primitive-recursive formula-negation token machine
Public local-check numeric graph
Accepted-code bounded modus ponens compiler
Accepted-code bounded conjunction compiler
ListedCertifiedFiniteConsistency(k)
PA consistency in the standard model
Consistency contradiction + exponent extraction
Bounded proof-calculus assembly (i)-(v)
Pudlak condition (i)
Pudlak condition (ii)
Pudlak condition (iii)
Pudlak condition (iv)
suitable numeration
PA computation-trace theorem
B. Decoder integrity（解码完整性）
decodeCompactTerm / Formula / Sequent
decodeCompactProof
decodePAAxiomCertificate / decodeStructuralValidityCertificate
decodeCompactPackedCertifiedPAProof
D. Explicit local costed verifier（显式带成本局部验证器）
bitStringEqTrace / natEqTrace
bitStringMemTrace / SubsetTrace / SetEqTrace
formulaEq/Mem/Subset/SetEqTrace
ListedCheckedPAProofTree
listedCertificateValidTrace
F. Active total bit-cost proof（当前总位成本证明）
decodeBinaryNatTrace_result
decodeBinaryNatTrace_success/failure_potential
OptionTrace + traceBind
decodeManyVectorTrace_potential
arithmeticFunc/Rel_decode_charge_le
decodeManyVectorTrace_result
decodeCompactTermTrace_parserCostBound
decodeCompactTermTrace_result
decodeCompactFormulaTrace_parserCostBound
decodeCompactFormulaTrace_result
decodeCompactListedProof_honestWeight_le
formulaEq/Mem/Subset/SetEqTrace_cost_le_polynomial
decodeCompactListedProofTrace_parserCostBound
PA/structural certificate decoder traces
decodeCompactListedProofTrace_result
transformation binary-code length bounds
10/10 listedCertificateValidTrace cost branches
decodePAAxiomCertificateTrace_parserCostBound
listedCompactCertifiedPAProofVerifierTrace_result
decodeCompactPackedListedCertifiedPAProof_honestWeight_le
packed proof/formula decoder cost bounds
axm comparator input
binaryFormulaNegTrace_result
guarded axiom trace
guardedListedCertificateValidTrace_jointCost_le
Success/failure total potential bound
VerifierBitCost(x,y) <= p(|x|+|y|)
图例（status / 状态）
Lean kernel checked
Active proof obligation
Pending root theorem
Axiom/dependency audit
Rejected route